

Database Technologies - Assignment 1

1. Create 'spj' database'. (Database for suppliers, parts and jobs)

```
[root@localhost>CREATE DATABASE spj;
Query OK, 1 row affected (0.004 sec)

[root@localhost>GRANT ALL ON spj.* TO 'D6_96342_Jagdish'@'localhost';
Query OK, 0 rows affected (0.001 sec)

[root@localhost>SHOW DATABASES
[    -> ;
+-----+
| Database |
+-----+
| classwork |
| hr        |
| information_schema |
| mysql     |
| performance_schema |
| sales     |
| spj       |
| sys       |
+-----+
8 rows in set (0.001 sec)
```

2. Use spj database.

```
D6_96342_Jagdish@localhost>USE spj
Database changed
D6_96342_Jagdish@localhost>CREATE TABLE S (`S#` CHAR(5) NOT NULL, Sname CHAR(20), status SMALLINT, city CHAR(15));
Query OK, 0 rows affected (0.016 sec)

D6_96342_Jagdish@localhost>DESCRIBE S;
+-----+-----+-----+-----+-----+-----+
| Field | Type   | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| S#    | char(5) | NO   |     | NULL    |       |
| Sname | char(20) | YES  |     | NULL    |       |
| status | smallint | YES  |     | NULL    |       |
| city  | char(15) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+
4 rows in set (0.004 sec)

D6_96342_Jagdish@localhost>
```

3. Create following tables in spj database.

CREATE TABLE S (S# CHAR(5) NOT NULL, Sname CHAR(20), status SMALLINT, city CHAR(15));

Note: S denotes the Suppliers Table.

```
D6_96342_Jagdish@localhost>CREATE TABLE S ( 'S#' CHAR(5) NOT NULL PRIMARY KEY, Sname CHAR(20), status SMALLINT, city CHAR(15) );
Query OK, 0 rows affected (0.009 sec)
```

```
D6_96342_Jagdish@localhost>DESCRIBE S;
```

Field	Type	Null	Key	Default	Extra
S#	char(5)	NO	PRI	NULL	
Sname	char(20)	YES		NULL	
status	smallint	YES		NULL	
city	char(15)	YES		NULL	

```
4 rows in set (0.003 sec)
```

```
D6_96342_Jagdish@localhost>
```

```
D6_96342_Jagdish@localhost>INSERT INTO S VALUES ('S1', 'Smith', 20, 'London'), ('S2', 'Jones', 10, 'Paris'), ('S3', 'Blake', 30, 'Paris'),
Query OK, 5 rows affected (0.003 sec), 'Adams', 30, 'Athens');
Records: 5 Duplicates: 0 Warnings: 0
```

```
D6_96342_Jagdish@localhost>SELECT * FROM S;
```

S#	Sname	status	city
S1	Smith	20	London
S2	Jones	10	Paris
S3	Blake	30	Paris
S4	Clark	20	London
S5	Adams	30	Athens

```
5 rows in set (0.001 sec)
```

```
D6_96342_Jagdish@localhost>
```

CREATE TABLE P (P# CHAR(6) NOT NULL, Pname CHAR(20), color CHAR(6), weight SMALLINT, city CHAR(15));

Note: P denotes the Parts table.

```
D6_96342_Jagdish@localhost>CREATE TABLE P ( `P#` CHAR(6) NOT NULL PRIMARY KEY, Pname CHAR(20), color CHAR(6), weight SMALLINT, city CHAR(15) );
Query OK, 0 rows affected (0.010 sec)

D6_96342_Jagdish@localhost>DESCRIBE P;
```

Field	Type	Null	Key	Default	Extra
P#	char(6)	NO	PRI	NULL	
Pname	char(20)	YES		NULL	
color	char(6)	YES		NULL	
weight	smallint	YES		NULL	
city	char(15)	YES		NULL	

```
5 rows in set (0.003 sec)
```

```
D6_96342_Jagdish@localhost>INSERT INTO P VALUES ('P1', 'Nut', 'Red', 12, 'London'), ('P2', 'Bolt', 'Green', 17, 'Paris'),
-> ('P3', 'Screw', 'Blue', 17, 'Rome'), ('P4', 'Screw', 'Red', 14, 'London'),
-> ('P5', 'Cam', 'Blue', 12, 'Paris'), ('P6', 'Cog', 'Red', 19, 'London');
Query OK, 6 rows affected (0.002 sec)
Records: 6 Duplicates: 0 Warnings: 0

D6_96342_Jagdish@localhost>SELECT * FROM P;
```

P#	Pname	color	weight	city
P1	Nut	Red	12	London
P2	Bolt	Green	17	Paris
P3	Screw	Blue	17	Rome
P4	Screw	Red	14	London
P5	Cam	Blue	12	Paris
P6	Cog	Red	19	London

```
6 rows in set (0.001 sec)
```

CREATE TABLE J (J# CHAR(4) NOT NULL, Jname CHAR(10), City CHAR(15));

Note: J denotes the Jobs table.

```
D6_96342_Jagdish@localhost>CREATE TABLE J ( `J#` CHAR(4) NOT NULL PRIMARY KEY, Jname CHAR(10), City CHAR(15) );
Query OK, 0 rows affected (0.009 sec)
```

```
D6_96342_Jagdish@localhost>DESCRIBE J;
```

Field	Type	Null	Key	Default	Extra
J#	char(4)	NO	PRI	NULL	
Jname	char(10)	YES		NULL	
City	char(15)	YES		NULL	

```
3 rows in set (0.002 sec)
```

```
D6_96342_Jagdish@localhost>INSERT INTO J VALUES ('J1', 'Sorter', 'Paris'), ('J2', 'Punch', 'Rome'),
-> ('J3', 'Reader', 'Athens'), ('J4', 'Console', 'Athens'), ('J5', 'Collator', 'London'),
-> ('J6', 'Terminal', 'Oslo'), ('J7', 'Tape', 'London');
```

```
Query OK, 7 rows affected (0.002 sec)
Records: 7 Duplicates: 0 Warnings: 0
```

```
D6_96342_Jagdish@localhost>DESCRIBE J;
```

Field	Type	Null	Key	Default	Extra
J#	char(4)	NO	PRI	NULL	
Jname	char(10)	YES		NULL	
City	char(15)	YES		NULL	

```
3 rows in set (0.002 sec)
```

```
D6_96342_Jagdish@localhost>
```

```
D6_96342_Jagdish@localhost>SELECT * FROM J;
```

J#	Jname	City
J1	Sorter	Paris
J2	Punch	Rome
J3	Reader	Athens
J4	Console	Athens
J5	Collator	London
J6	Terminal	Oslo
J7	Tape	London

```
7 rows in set (0.001 sec)
```

CREATE TABLE SP (S# CHAR(4) NOT NULL, P# CHAR(4) NOT NULL, J# CHAR(4) NOT NULL, QTY INT);

```
D6_96342_Jagdish@localhost>CREATE TABLE SP (`S#` CHAR(4) NOT NULL, `P#` CHAR(4) NOT NULL, `J#` CHAR(4) NOT NULL, QTY INT);
Query OK, 0 rows affected (0.009 sec)
```

```
D6_96342_Jagdish@localhost>DESCRIBE SP;
```

Field	Type	Null	Key	Default	Extra
S#	char(4)	NO		NULL	
P#	char(4)	NO		NULL	
J#	char(4)	NO		NULL	
QTY	int	YES		NULL	

```
4 rows in set (0.003 sec)
```

```
D6_96342_Jagdish@localhost>
```

```
D6_96342_Jagdish@localhost>INSERT INTO SP VALUES
```

```
-> ('S1','P1','J1',200), ('S1','P1','J4',700), ('S2','P3','J1',400), ('S2','P3','J2',200),
-> ('S2','P3','J3',200), ('S2','P3','J4',500), ('S2','P3','J5',600), ('S2','P3','J6',400),
-> ('S2','P3','J7',800), ('S2','P5','J2',100), ('S3','P3','J1',200), ('S3','P4','J2',500),
-> ('S4','P6','J3',300), ('S4','P6','J7',300), ('S5','P2','J2',200), ('S5','P2','J4',100),
-> ('S5','P5','J5',500), ('S5','P5','J7',100), ('S5','P6','J2',200), ('S5','P1','J4',100),
-> ('S5','P3','J4',200), ('S5','P4','J4',800), ('S5','P5','J4',400), ('S5','P6','J4',500);
```

```
Query OK, 24 rows affected (0.003 sec)
```

```
Records: 24 Duplicates: 0 Warnings: 0
```

```
D6_96342_Jagdish@localhost>SELECT * FROM SP;
```

S#	P#	J#	QTY
S1	P1	J1	200
S1	P1	J4	700
S2	P3	J1	400
S2	P3	J2	200
S2	P3	J3	200
S2	P3	J4	500
S2	P3	J5	600
S2	P3	J6	400
S2	P3	J7	800
S2	P5	J2	100
S3	P3	J1	200
S3	P4	J2	500
S4	P6	J3	300

NOTE – Database Creation and user Creation, assigning privileges is responsibility of Database Administrator (root)

1. Create 'hr' database. Import hr-db.sql.

```
[root@localhost>CREATE DATABASE hr;
Query OK, 1 row affected (0.003 sec)

[root@localhost>USE hr
Database changed
[root@localhost>SOURCE /Users/jagdishbutte/Desktop/Sunbeam/DBT/Assignment01/hr-db.sql
Query OK, 0 rows affected (0.000 sec)

Query OK, 0 rows affected (0.000 sec)

Query OK, 0 rows affected (0.000 sec)

Query OK, 0 rows affected (0.000 sec)

Query OK, 0 rows affected (0.000 sec)

Query OK, 0 rows affected (0.000 sec)

Query OK, 0 rows affected (0.000 sec)
```

2. Create 'sales' database. Import sales-db.sql.

```
[root@localhost>CREATE DATABASE sales;
Query OK, 1 row affected (0.003 sec)

[root@localhost>USE sales;
Database changed
[root@localhost>SOURCE /Users/jagdishbutte/Desktop/Sunbeam/DBT/Assignment01/sales-db.sql
Query OK, 0 rows affected, 1 warning (0.009 sec)

Query OK, 0 rows affected, 3 warnings (0.008 sec)

Query OK, 0 rows affected, 3 warnings (0.004 sec)

Query OK, 1 row affected (0.001 sec)

Query OK, 1 row affected (0.000 sec)

Query OK, 1 row affected (0.000 sec)

Query OK, 1 row affected (0.000 sec)

Query OK, 1 row affected (0.000 sec)

Query OK, 1 row affected (0.001 sec)
```

3. List all tables in hr database. Display contents of all tables (one by one).

```
D6_96342_Jagdish@localhost>USE hr;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A
```

Database changed

```
[D6_96342_Jagdish@localhost>SHOW TABLES;
```

```

+-----+
| Tables_in_hr |
+-----+
| countries     |
| departments   |
| employees     |
| job_history   |
| jobs          |
| locations     |
| regions       |

```

```
7 rows in set (0.002 sec)
```

```
[D6_96342_Jagdish@localhost>SELECT * FROM countries;
```

COUNTRY_ID	COUNTRY_NAME	REGION_ID
AR	Argentina	20
AU	Australia	40
BE	Belgium	10
BR	Brazil	20
CA	Canada	20
CH	Switzerland	10
CN	China	30
DE	Germany	10
DK	Denmark	10
EG	Egypt	50
FR	France	10
GB	United Kingdom of Great Britain and Nort	10
IL	Israel	30
IN	India	30
IT	Italy	10
JP	Japan	30
KW	Kuwait	30
ML	Malaysia	30
MX	Mexico	20
NG	Nigeria	50
NL	Netherlands	10
SG	Singapore	30
US	United States of America	20
ZM	Zambia	50

D6_96342_Jagdish@localhost>SELECT * FROM departments;

DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
10	Administration	200	1700
20	Marketing	201	1800
30	Purchasing	114	1700
40	Human Resources	203	2400
50	Shipping	121	1500
60	IT	103	1400
70	Public Relations	204	2700
80	Sales	145	2500
90	Executive	100	1700
100	Finance	108	1700
110	Accounting	205	1700
120	Treasury	NULL	1700
130	Corporate Tax	NULL	1700
140	Control And Credit	NULL	1700
150	Shareholder Services	NULL	1700
160	Benefits	NULL	1700
170	Manufacturing	NULL	1700
180	Construction	NULL	1700
190	Contracting	NULL	1700
200	Operations	NULL	1700
210	IT Support	NULL	1700
220	NOC	NULL	1700
230	IT Helpdesk	NULL	1700
240	Government Sales	NULL	1700

D6_96342_Jagdish@localhost>SELECT * FROM jobs;

JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
AC_ACCOUNT	Public Accountant	4200	9000
AC_MGR	Accounting Manager	8200	16000
AD_ASST	Administration Assistant	3000	6000
AD PRES	President	20080	40000
AD_VP	Administration Vice President	15000	30000
FI_ACCOUNT	Accountant	4200	9000
FI_MGR	Finance Manager	8200	16000
HR_REP	Human Resources Representative	4000	9000
IT_PROG	Programmer	4000	10000
MK_MAN	Marketing Manager	9000	15000
MK_REP	Marketing Representative	4000	9000
PR_REP	Public Relations Representative	4500	10500
PU_CLERK	Purchasing Clerk	2500	5500
PU_MAN	Purchasing Manager	8000	15000
SA_MAN	Sales Manager	10000	20080
SA_REP	Sales Representative	6000	12008
SH_CLERK	Shipping Clerk	2500	5500
ST_CLERK	Stock Clerk	2008	5000
ST_MAN	Stock Manager	5500	8500

19 rows in set (0.001 sec)

D6_96342_Jagdish@localhost>SELECT * FROM employees;

EMPLOYEE_ID	FIRST_NAME	LAST_NAME	EMAIL	PHONE_NUMBER	HIRE_DATE	JOB_ID	SALARY	COMMISSION_PCT	MANAGER_ID	DEPARTMENT_ID
100	Steven	King	SKING	1.515.555.0100	2013-06-17	AD_PRES	24000.00	NULL	NULL	90
101	Neena	Yang	NYANG	1.515.555.0101	2015-09-21	AD_VP	17000.00	NULL	100	90
102	Lex	Garcia	LGARCIA	1.515.555.0102	2011-01-13	AD_VP	17000.00	NULL	100	90
103	Alexander	James	AJAMES	1.590.555.0103	2016-01-03	IT_PROG	9000.00	NULL	102	60
104	Bruce	Miller	BMILLER	1.590.555.0104	2017-05-21	IT_PROG	6000.00	NULL	103	60
105	David	Williams	DWILLIAMS	1.590.555.0105	2015-06-25	IT_PROG	4800.00	NULL	103	60
106	Valli	Jackson	VJACKSON	1.590.555.0106	2016-02-05	IT_PROG	4800.00	NULL	103	60
107	Diana	Nguyen	DNGUYEN	1.590.555.0107	2017-02-07	IT_PROG	4200.00	NULL	103	60
108	Nancy	Gruenberg	NGRUENBE	1.515.555.0108	2012-08-17	FI_MGR	12000.00	NULL	101	100
109	Daniel	Faviet	DFAVIET	1.515.555.0109	2012-08-16	FI_ACCOUNT	9000.00	NULL	108	100
110	John	Chen	JCHEN	1.515.555.0110	2015-09-28	FI_ACCOUNT	8200.00	NULL	108	100
111	Ismail	Sciarra	ISCIARRA	1.515.555.0111	2015-09-30	FI_ACCOUNT	7700.00	NULL	108	100
112	Jose Manuel	Urman	JMURMAN	1.515.555.0112	2016-03-07	FI_ACCOUNT	7800.00	NULL	108	100
113	Luis	Popp	LPOPP	1.515.555.0113	2017-12-07	FI_ACCOUNT	6900.00	NULL	108	100
114	Den	Li	DLI	1.515.555.0114	2012-12-07	PU_MAN	11000.00	NULL	100	30
115	Alexander	Khoo	AKHOO	1.515.555.0115	2013-05-18	PU_CLERK	3100.00	NULL	114	30
116	Shelli	Baida	SBAIDA	1.515.555.0116	2015-12-24	PU_CLERK	2900.00	NULL	114	30
117	Sigal	Tobias	STOBIAS	1.515.555.0117	2015-07-24	PU_CLERK	2800.00	NULL	114	30
118	Guy	Himuro	GHIIMURO	1.515.555.0118	2016-11-15	PU_CLERK	2600.00	NULL	114	30
119	Karen	Colmenares	KCOLMENA	1.515.555.0119	2017-08-10	PU_CLERK	2500.00	NULL	114	30
120	Matthew	Weiss	MWEISS	1.650.555.0120	2014-07-18	ST_MAN	8000.00	NULL	100	50
121	Adam	Fripp	AFRIPP	1.650.555.0121	2015-04-10	ST_MAN	8200.00	NULL	100	50
122	Payam	Kaufling	PKAUFLIN	1.650.555.0122	2013-05-01	ST_MAN	7900.00	NULL	100	50
123	Shanta	Vollman	SVOLLMAN	1.650.555.0123	2015-10-10	ST_MAN	6500.00	NULL	100	50

D6_96342_Jagdish@localhost>SELECT * FROM jobs;

JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
AC_ACCOUNT	Public Accountant	4200	9000
AC_MGR	Accounting Manager	8200	16000
AD_ASST	Administration Assistant	3000	6000
AD_PRES	President	20080	40000
AD_VP	Administration Vice President	15000	30000
FI_ACCOUNT	Accountant	4200	9000
FI_MGR	Finance Manager	8200	16000
HR_REP	Human Resources Representative	4000	9000
IT_PROG	Programmer	4000	10000
MK_MAN	Marketing Manager	9000	15000
MK_REP	Marketing Representative	4000	9000
PR_REP	Public Relations Representative	4500	10500
PU_CLERK	Purchasing Clerk	2500	5500
PU_MAN	Purchasing Manager	8000	15000
SA_MAN	Sales Manager	10000	20080
SA_REP	Sales Representative	6000	12008
SH_CLERK	Shipping Clerk	2500	5500
ST_CLERK	Stock Clerk	2008	5000
ST_MAN	Stock Manager	5500	8500

19 rows in set (0.001 sec)

```
[D6_96342_Jagdish@localhost>SELECT * FROM locations;
```

LOCATION_ID	STREET_ADDRESS	POSTAL_CODE	CITY	STATE_PROVINCE	COUNTRY_ID
1000	1297 Via Cola di Rie	00989	Roma	NULL	IT
1100	93091 Calle della Testa	10934	Venice	NULL	IT
1200	2017 Shinjuku-ku	1689	Tokyo	Tokyo Prefecture	JP
1300	9450 Kamiya-cho	6823	Hiroshima	NULL	JP
1400	2014 Jabberwocky Rd	26192	Southlake	Texas	US
1500	2011 Interiors Blvd	99236	South San Francisco	California	US
1600	2007 Zagora St	50090	South Brunswick	New Jersey	US
1700	2004 Charade Rd	98199	Seattle	Washington	US
1800	147 Spadina Ave	MSV 2L7	Toronto	Ontario	CA
1900	6092 Boxwood St	YSW 9T2	Whitehorse	Yukon	CA
2000	40-5-12 Laogianggen	190518	Beijing	NULL	CN
2100	1298 Vileparle (E)	40231	Bombay	Maharashtra	IN
2200	12-98 Victoria Street	2901	Sydney	New South Wales	AU
2300	198 Clementi North	540198	Singapore	NULL	SG
2400	8204 Arthur St	NULL	London	NULL	GB
2500	Magdalen Centre, The Oxford Science Park	OX9 9ZB	Oxford	Oxford	GB
2600	9702 Chester Road	09629850293	Stretford	Manchester	GB
2700	Schwanthalerstr. 7031	80925	Munich	Bavaria	DE
2800	Rua Frei Caneca 1360	01307-002	Sao Paulo	Sao Paulo	BR
2900	20 Rue des Corps-Saints	1730	Geneva	Geneve	CH
3000	Murtenstrasse 921	3095	Bern	BE	CH
3100	Pieter Breughelstraat 837	3029SK	Utrecht	Utrecht	NL
3200	Mariano Escobedo 9991	11932	Mexico City	Distrito Federal	MX

```
[D6_96342_Jagdish@localhost>SELECT * FROM regions;
```

REGION_ID	REGION_NAME
50	Africa
20	Americas
30	Asia
10	Europe
40	Oceania

```
5 rows in set (0.000 sec)
```

4. List all tables in sales database. Display contents of all tables (one by one).

```
[D6_96342_Jagdish@localhost>SHOW TABLES;
```

Tables_in_sales
customers
orders
salespeople

3 rows in set (0.002 sec)

```
D6_96342_Jagdish@localhost>
```

```
[D6_96342_Jagdish@localhost>SELECT * FROM customers;
```

cnum	cname	city	rating	snum
2001	Hoffman	London	100	1001
2002	Giovanni	Rome	200	1003
2003	Liu	San Jose	200	1002
2004	Grass	Berlin	300	1002
2006	Clemens	London	100	1001
2008	Cisneros	San Jose	300	1007
2007	Pereira	Rome	100	1004

7 rows in set (0.000 sec)

```
[D6_96342_Jagdish@localhost>SELECT * FROM orders;
```

onum	amt	odate	cnum	snum
3001	18.69	1990-10-03	2008	1007
3003	767.19	1990-10-03	2001	1001
3002	1900.10	1990-10-03	2007	1004
3005	5160.45	1990-10-03	2003	1002
3006	1098.16	1990-10-03	2008	1007
3009	1713.23	1990-10-04	2002	1003
3007	75.75	1990-10-04	2004	1002
3008	4723.00	1990-10-04	2006	1001
3010	309.95	1990-10-04	2004	1002
3011	9891.88	1990-10-04	2006	1001

```
10 rows in set (0.001 sec)
```

```
[D6_96342_Jagdish@localhost>SELECT * FROM salespeople;
```

snum	sname	city	comm
1001	Peel	London	0.12
1002	Serres	San Jose	0.13
1004	Motika	London	0.11
1007	Rifkin	Barcelona	0.15
1003	Axelrod	New York	0.10

```
5 rows in set (0.001 sec)
```

Answer the following questions (by observations only – not queries).

1. How many orders data is present in the database?

ANS -> 10

2. How many customers are present in the database?

ANS -> 7

3. How many salespeople are present in the database?

ANS -> 5

4. On which date, order with highest amount is placed?

ANS -> 1990-10-04

5. For which order salesman got maximum commission? Hint: you need to refer data of two tables.

ANS -> Order number 3011

6. Which salesmen are handling more than two customers? Hint: you need to refer data of two tables.

ANS -> Peel and Serres

7. Which city have single customer?

ANS -> Berlin

8. Which city has multiple salespeople?

ANS -> London

9. What is name of customer and salesman of the maximum amount order?

ANS -> Customer - Clemens, Salesmen - Peel

10. Study relationship between the tables in sales database.

ANS -> All 3 tables have a common field names sname. Customer and Orders table have 2 common fields, sname and cname.